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Micro light emitting diode structure and micro light emitting diode display device

Abstract

A micro light emitting diode structure including an epitaxial structure, an electrode layer, a barrier layer and a bonding layer is provided. The epitaxial structure has a surface. The electrode layer is disposed on the surface of the epitaxial structure. The barrier layer is disposed on the electrode layer. The bonding layer is disposed on the barrier layer and away from the electrode layer.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation-in-part application of and claims the priority benefit of U.S. application Ser. No. 17/748,049, filed on May 19, 2022, now pending, which claims the priority benefit of Taiwan application serial no. 111115561, filed on Apr. 25, 2022. The entirety of each of the above-mentioned patent applications is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND
Technical Field
(1) The invention relates to a light emitting structure and a light emitting device, and particularly relates to a micro light emitting diode structure and a micro light emitting diode display device.
Description of Related Art
(2) In an existing solder bump process, an orthogonal projection area of a barrier layer on an epitaxial structure is equal to an orthogonal projection area of an electrode layer on the epitaxial structure, so that when subsequent transfer and bonding with a display back plate is implemented to perform a reflow process, a high temperature may easily lead to formation of a metal eutectic between the solder bump and the electrode layer, thereby affecting electrical properties and structural reliability of a micro light emitting diode display device.
SUMMARY
(3) The invention is directed to a micro light emitting diode structure, which has better electrical properties and structural reliability.
(4) The invention is directed to a micro light emitting diode display device including the aforementioned micro light emitting diode structure, and has a better display yield.
(5) The invention provides a micro light emitting diode structure including an epitaxial structure, an electrode layer, a barrier layer and a bonding layer is provided. The epitaxial structure has a surface. The electrode layer is disposed on the surface of the epitaxial structure. The barrier layer is disposed on the electrode layer. The bonding layer is disposed on the barrier layer and away from the electrode layer.
(6) The invention provides a micro light emitting diode display device including a display substrate and a plurality of micro light emitting diode structures. The micro light emitting diode structures are disposed on the display substrate and respectively include an epitaxial structure, an electrode layer, a barrier layer and a bonding layer. The epitaxial structure has a surface. The electrode layer is disposed on the surface of the epitaxial structure and exposes an electrode surface. The barrier layer is disposed on the electrode layer and completely covers the electrode surface. The bonding layer is disposed on the barrier layer. The bonding layer is bonded to the corresponding pads, so that the micro light-emitting diode structures are electrically connected to the display substrate.
(7) Based on the above description, in the design of the micro light emitting diode structure of the invention, the barrier layer is disposed on the electrode layer, and the bonding layer is disposed on the barrier layer and away from the electrode layer. Namely, in the invention, the barrier layer is used to prevent the eutectic problem between the electrode layer and the bonding layer, and can make the bonding layer disposed on the barrier layer have better adhesion. Therefore, the micro light emitting diode structure of the invention may have better electrical properties and structural reliability, and the micro light emitting diode display device using the micro light emitting diode structures of the invention may have better display yield.

(8) To make the aforementioned more comprehensible, several embodiments accompanied with drawings are described in detail as follows.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings are included to provide a further understanding of the invention, and are incorporated in and constitute a part of this specification. The drawings illustrate embodiments of the invention and, together with the description, serve to explain the principles of the invention.

(2) FIG. **1**A is a schematic cross-sectional view of a micro light emitting diode structure according to an embodiment of the invention.

(3) FIG. **1**B is a schematic top view of the micro light emitting diode structure of FIG. **1**A.

(4) FIG. **1**C is a schematic cross-sectional view of a micro light emitting diode structure according to another embodiment of the invention.

(5) FIG. **2** is a schematic cross-sectional view of a micro light emitting diode structure according to another embodiment of the invention.

(6) FIG. **3** is a schematic cross-sectional view of a micro light emitting diode structure according to another embodiment of the invention.

(7) FIG. **4** is a schematic cross-sectional view of a micro light emitting diode structure according to another embodiment of the invention.

(8) FIG. **5** is a schematic cross-sectional view of a micro light emitting diode structure according to another embodiment of the invention.

(9) FIG. **6** is a schematic cross-sectional view of a micro light emitting diode structure according to another embodiment of the invention.

(10) FIG. **7** is a schematic top view of a micro light emitting diode structure according to another embodiment of the invention.

(11) FIG. **8** is a schematic top view of a micro light emitting diode structure according to another embodiment of the invention.

(12) FIG. **9** is a schematic cross-sectional view of a micro light emitting diode display device according to an embodiment of the invention.

(13) FIG. **10** is a schematic top view of a micro light emitting diode structure according to another embodiment of the invention.

(14) FIG. **11** is a schematic top view of a micro light emitting diode structure according to another embodiment of the invention.

(15) FIG. **12** is a schematic top view of a micro light emitting diode structure according to another embodiment of the invention.

(16) FIG. **13** is a schematic top view of a micro light emitting diode structure according to another embodiment of the invention.

(17) FIG. **14** is a schematic top view of a micro light emitting diode structure according to another embodiment of the invention.

(18) FIG. **15** is a schematic cross-sectional view of a micro light emitting diode display device according to another embodiment of the invention.

DESCRIPTION OF THE EMBODIMENTS

(19) FIG. **1**A is a schematic cross-sectional view of a micro light emitting diode structure according to an embodiment of the invention. FIG. **1**B is a schematic top view of the micro light emitting diode structure of FIG. **1**A. Referring to FIG. **1**A first, in the embodiment, the micro light emitting diode structure **100***a* includes an epitaxial structure **110***a*, an electrode layer **120***a* and a barrier layer **130***a*. The epitaxial structure **110***a* has a surface S**1**. The electrode layer **120***a* is disposed on

the surface S**1** of the epitaxial structure **110**_a_. The barrier layer **130**_a_ is disposed on the electrode layer **120**_a_. An orthogonal projection area of the barrier layer **130**_a_ on the epitaxial structure **110**_a_ is greater than and covers an orthogonal projection area of the electrode layer **120**_a_ on the epitaxial structure **110**_a_. Namely, the barrier layer **130**_a_ may contact the electrode layer **120**_a_ avoid a eutectic phenomenon generated during a subsequent reflow process, which ensures that the micro light emitting diode structure **100**_a_ of the embodiment has better electrical properties and structural reliability.
(20) In detail, in the embodiment, the epitaxial structure **110**_a_ includes a first-type semiconductor layer **112**_a_, a light emitting layer **114**_a_, a second-type semiconductor layer **116**_a_, and a through hole **115**_a_. The light emitting layer **114**_a_ is located between the first-type semiconductor layer **112**_a_ and the second-type semiconductor layer **116**_a_. The through hole **115**_a_ penetrates through the first-type semiconductor layer **112**_a_, the light emitting layer **114**_a_ and a part of the second-type semiconductor layer **116**_a_. One of the first-type semiconductor layer **112**_a_ and the second-type semiconductor layer **116**_a_ is a P-type semiconductor layer, and the other one of the first-type semiconductor layer **112**_a_ and the second-type semiconductor layer **116**_a_ is an N-type semiconductor layer. In addition, the micro light emitting diode structure **100**_a_ of the embodiment further includes an insulating layer **140**_a_ disposed on the epitaxial structure **110**_a_ and covering the surface S**1** and a peripheral surface S**2** of the epitaxial structure **110**_a_. The insulating layer **140**_a_ has a first opening **142**_a_ exposing the first-type semiconductor layer **112**_a_ and a second opening **144**_a_ exposing the second-type semiconductor layer **116**_a_, where the insulating layer **140**_a_ extends to cover an inner wall of the through hole **115**_a_. Here, a cross-sectional profile of the epitaxial structure **110**_a_ is, for example, a trapezoidal profile, where a side surface of the first type semiconductor layer **112**_a_, a side surface of the light emitting layer **114**_a_ and a side surface of the second type semiconductor layer **116**_a_ in the epitaxial structure **110**_a_ may be connected and extended obliquely in two stages, and the insulating layer **140**_a_ is disposed on the epitaxial structure **110**_a_ along the profile of the epitaxial structure **110**_a_ to achieve better yield.
(21) Moreover, the electrode layer **120**_a_ in the embodiment includes a first electrode **122**_a_ and a second electrode **124**_a_. The first electrode **122**_a_ is disposed on the insulating layer **140**_a_ and extends into the first opening **142**_a_ and is electrically connected to the first-type semiconductor layer **112**_a_. The second electrode **124**_a_ is disposed on the insulating layer **140**_a_ and extends into the second opening **144**_a_ and is electrically connected to the second-type semiconductor layer **116**_a_. The electrode layer **120**_a_ is, for example, a multi-layer film structure, where a material of the electrode layer **120**_a_ is, for example, copper, aluminum, platinum, titanium, gold, silver, chromium, or a combination thereof, but the invention is not limited thereto. Here, the first electrode **122**_a_ and the second electrode **124**_a_ are both located on the same side of the epitaxial structure **110**_a_, which means that the micro light emitting diode structure **100**_a_ of the embodiment is embodied as a flip-chip micro light emitting diode structure.
(22) In addition, the barrier layer **130**_a_ of the embodiment includes a first barrier portion **132**_a_ and a second barrier portion **134**_a_. The first barrier portion **132**_a_ covers the first electrode **122**_a_ and contacts the insulating layer **140**_a_ and the first electrode **122**_a_. The second barrier portion **134**_a_ covers the second electrode **124**_a_ and contacts the insulating layer **140**_a_ and the second electrode **124**_a_. Here, the first barrier portion **132**_a_ and the second barrier portion **134**_a_ have flat surfaces on the side away from the first electrode **122**_a_ and the second electrode **124**_a_.
(23) Preferably, in the embodiment, the orthogonal projection area of the electrode layer **120**_a_ on the epitaxial structure **110**_a_ and the orthogonal projection area of the barrier layer **130**_a_ on the epitaxial structure **110**_a_ are preferably less than or equal to 0.95 and greater than or equal to 0.5, which may ensure that the barrier layer **130**_a_ completely covers the electrode layer **120**_a_ to provide sufficient protection during a subsequent bonding process. In addition, a ratio of a thickness T of the barrier layer **130**_a_ to a thickness T**1** of the epitaxial structure **110**_a_ in this embodiment is preferably less than or equal to 0.1. If the ratio is above 0.1, it will affect subsequent electrical

contact properties. Referring to FIG. **1**A again, there is a gap G between a first edge E**1** of the barrier layer **130***a* and a second edge E**2** of the corresponding electrode layer **120***a* (for example, a gap between the second barrier portion **134***a* and the second electrode **124***a*), and a ratio of the gap G to a width W of the barrier layer **130***a* is greater than or equal to 0.05 and less than or equal to 0.5, where the gap G is, for example, greater than or equal to 0.3 μm and less than or equal to 10 μm, which may ensure that the barrier layer **130***a* completely covers the electrode layer **120***a* to provide sufficient protection. In addition, there is a horizontal distance H between the first barrier portion **132***a* and the second barrier portion **134***a*, and a ratio of the horizontal distance H to a width L of the epitaxial structure **110***a* in the same direction is greater than or equal to 0.01 and less than or equal to 0.3, where the horizontal distance H, for example, is between 0.3 μm and 15 μm, which may avoid a phenomenon of short circuit during the subsequent bonding process.

(24) In addition, referring to FIG. **1**A and FIG. **1**B again, in the embodiment, the micro light emitting diode structure **100***a* further includes a bonding layer **150***a* disposed on the barrier layer **130***a*. Here, an orthogonal projection area of the bonding layer **150***a* on the epitaxial structure **110***a* is, for example, equal to the orthogonal projection area of the barrier layer **130***a* on the epitaxial structure **110***a*. Namely, the bonding layer **150***a* and the barrier layer **130***a* may be operated with a same photolithography process, which may effectively reduce the production cost. Moreover, in the embodiment, a ratio of the thickness T of the barrier layer **130***a* to a thickness T**2** of the bonding layer **150***a* is less than or equal to 0.2, preferably, 0.03 to 0.2. In addition, a melting point of the barrier layer **130***a* is greater than a melting point of the bonding layer **150***a*, so as to effectively block the bonding layer **150***a* and the electrode layer **120***a*. A material of the bonding layer **150***a* is, for example, tin or tin alloy, and a material of the barrier layer **130***a* is, for example, nickel, platinum, titanium-tungsten or tungsten, but the invention is not limited thereto.

(25) In brief, in the embodiment, the orthogonal projection area of the barrier layer **130***a* on the epitaxial structure **110***a* is larger than and covers the orthogonal projection area of the electrode layer **120***a* on the epitaxial structure **110***a*. Namely, in the embodiment, the barrier layer **130***a* is used to contact the electrode layer **120***a*, thereby effectively blocking the bonding layer **150***a* and the electrode layer **120***a*, so as to avoid a problem that the bonding layer **150***a* and the electrode layer **120***a* form eutectic during a subsequent reflow process. Therefore, the micro light emitting diode structure **100***a* of the embodiment may have better electrical properties and structural reliability. It should be noted that, as shown in FIG. **1**C, in the micro light emitting diode structure **100***b*, the bonding layer **150***b* may also be embodied as two solder balls.

(26) It should be noted that reference numbers of the components and a part of contents of the aforementioned embodiment are also used in the following embodiment, where the same reference numbers denote the same or like components, and descriptions of the same technical contents are omitted. The aforementioned embodiment may be referred for descriptions of the omitted parts, and detailed descriptions thereof are not repeated in the following embodiment.

(27) FIG. **2** is a schematic cross-sectional view of a micro light emitting diode structure according to another embodiment of the invention. Referring to FIG. **1**A and FIG. **2** at the same time, a micro light emitting diode structure **100***c* of the embodiment is similar to the micro light emitting diode structure **100***a* of FIG. **1**A, and a difference there between is that in the embodiment, a first electrode **122***c* of an electrode layer **120***c* has a first groove **123**, and a second electrode **124***c* has a second groove **125**. A first barrier portion **132***c* of a barrier layer **130***c* extends into the first groove **123** and is conformally disposed with the first electrode **122***c*. Namely, the first barrier portion **132***c* also has a groove **133**. A second barrier portion **134***c* of the barrier layer **130***c* extends into the second groove **125** and is conformally disposed with the second electrode **124***c*. Namely, the second barrier portion **134***c* also has a groove **135**. The bonding layer **150***a* is disposed on the first barrier portion **132***c* and the second barrier portion **134***c*, and fills the groove **133** of the first barrier portion **132***c* and the groove **135** of the second barrier portion **134***c*, so as to form an accommodating space through the groove **133** and the groove **135** during the subsequent bonding

process, which may effectively avoid the contact between the electrode layer **120***c* and the barrier layer **130***c*.
(28) FIG. **3** is a schematic cross-sectional view of a micro light emitting diode structure according to another embodiment of the invention. Referring to FIG. **1**A and FIG. **3** at the same time, a micro light emitting diode structure **100***d* of the embodiment is similar to the micro light emitting diode structure **100***a* of FIG. **1**A, and a difference there between is that in the embodiment, a pattern of an epitaxial structure **110***d* is different from that of the epitaxial structure **110***a*. In detail, in the embodiment, a first-type semiconductor layer **112***d*, a light emitting layer **114***d* and a first portion **116***d***1** of a second-type semiconductor layer **116***d* form a mesa M. A second portion **116***d***2** of the second-type semiconductor layer **116***d* forms a base B with respect to the mesa M. A first opening **142***d* of an insulating layer **140***d* is located on the mesa M, and a second opening **144***d* exposes the second portion **116***d***2** of the second-type semiconductor layer **116***d* and is located on the base B. To be specific, the micro light emitting diode structure **100***d* is, for example, a horizontal micro light emitting diode structure.
(29) FIG. **4** is a schematic cross-sectional view of a micro light emitting diode structure according to another embodiment of the invention. FIG. **1**A and FIG. **4** at the same time, a micro light emitting diode structure **100***e* of the embodiment is similar to the micro light emitting diode structure **100***a* of FIG. **1**A, and a difference there between is that in the embodiment, a barrier layer **130***e* at least contacts a part of a peripheral surface of the electrode layer **120***a*. In detail, a first barrier portion **132***e* contacts a part of a peripheral surface S**3** of the first electrode **122***a*, and a second barrier portion **134***e* contacts a part of a peripheral surface S**4** of the second electrode **124***a*. Namely, the peripheral surface S**3** of the first electrode **122***a* and the peripheral surface S**4** of the second electrode **124***a* are not completely covered by the first barrier portion **132***e* and the second barrier portion **134***e*, so that there is more allowable space in the subsequent bonding process, and the barrier layer **130***e* may be deformed to cover the peripheral surface S**3** of the first electrode **122***a* and the peripheral surface S**4** of the second electrode **124***a* at the same time under high pressure and high temperature, so as to avoid short circuit caused by overflow bonding.
(30) FIG. **5** is a schematic cross-sectional view of a micro light emitting diode structure according to another embodiment of the invention. Referring to FIG. **1**A and FIG. **5** at the same time, a micro light emitting diode structure **100***f* of the embodiment is similar to the micro light emitting diode structure **100***a* of FIG. **1**A, and a difference there between is that in the embodiment, a barrier layer **130***f* contacts a part of a peripheral surface S**5** of a bonding layer **150***f*. In detail, a first barrier portion **132***f* and a second barrier portion **134***f* contact a part of the peripheral surface S**5** of the corresponding bonding layer **150***f*, so as to prevent the bonding layer **150***f* from overflowing after the reflow process.
(31) FIG. **6** is a schematic cross-sectional view of a micro light emitting diode structure according to another embodiment of the invention. Referring to FIG. **1**A and FIG. **6** at the same time, a micro light emitting diode structure **100***g* of the embodiment is similar to the micro light emitting diode structure **100***a* of FIG. **1**A, and a difference there between is that in the embodiment, an orthogonal projection area of a bonding layer **150***g* on the epitaxial structure **110***a* is larger than an orthogonal projection area of the barrier layer **130***a* on the epitaxial structure **110***a*. Namely, different processes may be adopted to fabricate the bonding layer **150***g* and the barrier layer **130***a*, which not only effectively blocks the bonding layer **150***a* and the electrode layer **120***a*, but also increases a bonding area when the bonding layer **150***a* is subsequently bonded to pads of a display back plate.
(32) FIG. **7** is a schematic top view of a micro light emitting diode structure according to another embodiment of the invention. Referring to FIG. **1**B and FIG. **7** at the same time, a micro light emitting diode structure **100***i* of the embodiment is similar to the micro light emitting diode structure **100***a* of FIG. **1**B, and a difference there between is that in the embodiment, an orthogonal projection area of a bonding layer **150***i* on the epitaxial structure **110***a* is smaller than an orthogonal projection area of a barrier layer **130***i* on the epitaxial structure **110***a* and larger than an orthogonal

projection area of the electrode layer **120***a* on the epitaxial structure **110***a*. Namely, the orthogonal projection area of the bonding layer **150***i* on the epitaxial structure **110***a* is smaller than the orthogonal projection area of the corresponding first barrier portion **132***i* on the epitaxial structure **110***a* and larger than the orthogonal projection area of the first electrode **122***a* on the epitaxial structure **110***a*. The orthogonal projection area of the bonding layer **150***i* on the epitaxial structure **110***a* is smaller than the orthogonal projection area of the corresponding second barrier portion **134***i* on the epitaxial structure **110***a* and larger than the orthogonal projection area of the second electrode **124***a* on the epitaxial structure **110***a*. Preferably, a ratio of the orthogonal projection area of the bonding layer **150***i* on the epitaxial structure **110***a* to the orthogonal projection area of the barrier layer **130***i* on the epitaxial structure **110***a* is, for example, less than or equal to 0.9 and greater than or equal to 0.5, and if the above ratio is less than 0.5, a bonding strength of the subsequently bonded bonding layer **150***i* may be insufficient, and a ratio of the orthogonal projection area of the bonding layer **150***i* on the epitaxial structure **110***a* to the orthogonal projection area of the electrode layer **120***a* on the epitaxial structure **110***a* is, for example, greater than 1 and less than or equal to 1.5.
(33) Since the orthogonal projection area of the bonding layer **150***i* on the epitaxial structure **110***a* is smaller than the orthogonal projection area of the barrier layer **130***i* on the epitaxial structure **110***a*. i.e., the bonding layer **150***i* is retracted by a distance relative to the barrier layer **130***i*, a risk of overflow of the bonding layer **150***i* during the subsequent reflow process is effectively reduced. As shown in FIG. **8**, an orthogonal projection area of a bonding layer **150***j* on the epitaxial structure **110***a* is smaller than an orthogonal projection area of a barrier layer **130***j* on the epitaxial structure **110***a* and an orthogonal projection area of the electrode layer **120***a* on the epitaxial structure **110***a*, when the applied micro light emitting diode structure **100***j* is smaller, for example, smaller than or equal to 15 μm, the bonding layer **150***j* and the electrode layer **120***a* may be effectively blocked.
(34) FIG. **9** is a schematic cross-sectional view of a micro light emitting diode display device according to an embodiment of the invention. Referring to FIG. **9**, in the embodiment, the micro light emitting diode display device **10** includes a display substrate **20** and, for example, the micro light emitting diode structure **100***b* shown in FIG. **1**C, where the micro light emitting diode structure **100***b* is disposed on the display substrate **20**. In detail, in the embodiment, the display substrate **20** includes a plurality of pads **22**, and the bonding layer **150***b* is bonded to the pads **22**, so that the micro light emitting diode structure **100***b* is disposed on the display substrate **20** and is electrically connected to the display substrate **20**. An orthogonal projection area of the barrier layer **130***a* on the display substrate **20** is larger than and covers an orthogonal projection area of the electrode layer **120***a* on the display substrate **20**, which may effectively avoid a problem that the bonding layer **150***b* overflows to form a eutectic with the first electrode **122***a* and the second electrode **124***a* of the electrode layer **120***a* due to a high temperature and high pressure process of eutectic bonding when the micro light emitting diode structure **100***b* is bonded to the display substrate **20**. Here, the orthogonal projection area of the barrier layer **130***a* on the display substrate **20** is larger than the orthogonal projection area of the electrode layer **120***a* on the display substrate **20** and less than or equal to an orthogonal projection area of the pad **22** on the display substrate **20**. The orthogonal projection area of the bonding layer **150***b* on the display substrate **20** is greater than or equal to the orthogonal projection area of the barrier layer **130***a* on the display substrate **20**, which may effectively protect the bonding layer **150***b* from overflowing when the micro light emitting diode structure **100***b* is bonded to the display substrate **20**, and meanwhile achieves better bonding through a larger pad area and bonding layer area, when the applied micro light emitting diode structure **100***b* is larger, for example, larger than or equal to 15 μm. Therefore, the micro light emitting diode display device **10** of the embodiment may have a better display yield.
(35) It should be noted that in other embodiments that are not shown, the micro light emitting diode display device may include at least any one of the micro light emitting diode structures **100***a*, **100***b*, **100***c*, **100***d*, **100***e*, **100***f*, **100***g*, **100***i*, **100***j* according to actual requirements, which is not limited by

the invention. Namely, the number of the micro light emitting diode structures may be one or plural, and the micro light emitting diode structures may be the same structure or different structures, which may be selected according to actual requirements. In addition, the micro light emitting diode structure may be a red micro light emitting diode structure, a blue micro light emitting diode structure or a green micro light emitting diode structure. Moreover, the display substrate **20** of the embodiment may be, for example, a complementary metal-oxide-semiconductor (CMOS) substrate, a liquid crystal on silicon (LCOS) substrate, a thin film transistor (TFT) substrate or other substrates with working circuits, which are not limited by the invention.
(36) FIG. **10** is a schematic top view of a micro light emitting diode structure according to another embodiment of the invention. Referring to FIG. **10**, in the embodiment, the micro light emitting diode structure **200***a* includes an epitaxial structure **210**, an electrode layer **220***a*, a barrier layer **230***a* and a bonding layer **250***a*. The epitaxial structure **210** has a surface S**1**′. The electrode layer **220***a* is disposed on the surface S**1**′ of the epitaxial structure **210**. The barrier layer **230***a* is disposed on the electrode layer **220***a*. The bonding layer **250***a* is disposed on the barrier layer **230***a* and away from the electrode layer **220***a*. Namely, the barrier layer **230***a* is used to prevent the eutectic problem between the electrode layer **220***a* and the bonding layer **250***a*, and can make the bonding layer **250***a* disposed on the barrier layer **230***a* have better adhesion. Therefore, the micro light emitting diode structure **200***a* of the embodiment may have better electrical properties and structural reliability.
(37) In detail, in the embodiment, the epitaxial structure **210** includes a first-type semiconductor layer **212**, a light emitting layer **214**, a second-type semiconductor layer **216**, and a through hole **215**. The light emitting layer **214** is located between the first-type semiconductor layer **212** and the second-type semiconductor layer **216**. The through hole **215** penetrates through the first-type semiconductor layer **212**, the light emitting layer **214** and a part of the second-type semiconductor layer **216**. One of the first-type semiconductor layer **212** and the second-type semiconductor layer **216** is a P-type semiconductor layer, and the other one of the first-type semiconductor layer **212** and the second-type semiconductor layer **216** is an N-type semiconductor layer. In addition, the micro light emitting diode structure **200***a* of the embodiment further includes an insulating layer **240***a* disposed on the epitaxial structure **210** and covering the surface S**1**′ and a peripheral surface S**2**′ of the epitaxial structure **210**. The insulating layer **240***a* has a first opening **242***a* exposing the first-type semiconductor layer **212** and a second opening **244***a* exposing the second-type semiconductor layer **216**, where the insulating layer **240***a* extends to cover an inner wall of the through hole **215**. Here, a cross-sectional profile of the epitaxial structure **210** is, for example, a trapezoidal profile, where a side surface of the first type semiconductor layer **212**, a side surface of the light emitting layer **214** and a side surface of the second type semiconductor layer **216** in the epitaxial structure **210** may be connected and extended obliquely in two stages, and the insulating layer **240***a* is disposed on the epitaxial structure **210** along the profile of the epitaxial structure **210** to achieve better yield. The insulating layer **240***a* can be, for example, a distributed bragg reflector.
(38) Moreover, the electrode layer **220***a* is disposed on the epitaxial structure **210***a* and exposes a surface S**3**′, and the barrier layer **230***a* completely covers the surface S**3**′. The electrode layer **220***a* in the embodiment includes a first electrode **222***a* and a second electrode **224***a*. One of first electrode **222***a* and the second electrode **224***a* is a P-type electrode, and the other one of the first electrode **222***a* and the second electrode **224***a* is an N-type electrode. The first electrode **222***a* is disposed in the first opening **242***a* of the insulating layer **240***a* and electrically connected with the first-type semiconductor layer **212**. The second electrode **224***a* is disposed in the second opening **244***a* of the insulating layer **240***a* and electrically connected with the second-type semiconductor layer **216**. Here, the first electrode **222***a* and the second electrode **224***a* are both located on the same side of the epitaxial structure **210**, which means that the micro light emitting diode structure **200***a* of the embodiment is embodied as a flip-chip micro light emitting diode structure.
(39) In addition, the barrier layer **230***a* of the embodiment includes a first barrier portion **232***a* and

a second barrier portion **234***a*. The first barrier portion **232***a* covers the first electrode **222***a* and contacts the insulating layer **240***a* and the first electrode **222***a*. The second barrier portion **234***a* covers the second electrode **224***a* and contacts the insulating layer **240***a* and the second electrode **224***a*. Herein, ratios of orthographic projections of the first electrode **222***a* and the second electrode **224***a* on the epitaxial structure **210** to orthographic projections of the first barrier portion **232***a* and the second barrier portion **234***a* on the epitaxial structure **210** are respectively greater than or equal to 0.25 and less than or equal to 0.95. A material of the barrier layer **230***a* is, for example, nickel, platinum, titanium-tungsten or tungsten, but the invention is not limited thereto.

(40) Referring to FIG. **10** again, in the embodiment, the bonding layer **250***a* has an upper surface **252***a*, and the upper surface **252***a* is a convex curved surface. A curvature of the upper surface **252***a* gradually decreases from a portion close to the barrier layer **250***a* to another portion away from the barrier layer **250***a*. Here, in cross-section, the outline of the bonding layer **250***a* is a semicircle, but the invention is not limited thereto. Preferably, the bonding layer **250***a* has a maximum thickness L**1** at a center and an edge thickness L**2** at an edge, and the maximum thickness L**1** is greater than twice the edge thickness L**2**. Furthermore, an orthographic projection of the bonding layer **250***a* on the epitaxial structure **210** is equal to and covers an orthographic projection of the barrier layer **230***a* on the epitaxial structure **210**, and a ration of the orthographic projection of the bonding layer **250***a* to the orthographic projection of the barrier layer **230***a* is equal to 1. There is a distance D between an edge of the bonding layer **250***a* and an edge of the electrode layer **220***a*, and a ration of an orthographic projection of the distance D on the epitaxial structure **210** to an orthographic projection of the electrode layer **220***a* on the epitaxial structure **210** is greater than or equal to 0.2. Here, the distance D is, for example, greater than 1 micrometer. A material of the bonding layer **250***a* is, for example, tin or tin alloy, but the invention is not limited thereto.

(41) In brief, in the embodiment, the barrier layer **230***a* is disposed on the electrode layer **220***a*, and the bonding layer **250***a* is disposed on the barrier layer **230***a* and away from the electrode layer **220***a*. Namely, the barrier layer **230***a* is used to prevent the eutectic problem between the electrode layer **220***a* and the bonding layer **250***a*, and can make the bonding layer **250***a* disposed on the barrier layer **230***a* have better adhesion. Therefore, the micro light emitting diode structure **200***a* of the embodiment may have better electrical properties and structural reliability.

(42) FIG. **11** is a schematic top view of a micro light emitting diode structure according to another embodiment of the invention. Referring to FIG. **10** and FIG. **11** at the same time, a micro light emitting diode structure **200***b* of the embodiment is similar to the micro light emitting diode structure **200***a* of FIG. **10**, and a difference there between is that in the embodiment, the bonding layer **250***b* has an upper surface **252***b*, and the upper surface **252***b* is a convex curved surface. Here, in cross-section, the outline of the bonding layer **250***a* is a spherical, but the invention is not limited thereto.

(43) FIG. **12** is a schematic top view of a micro light emitting diode structure according to another embodiment of the invention. Referring to FIG. **10** and FIG. **12** at the same time, a micro light emitting diode structure **200***c* of the embodiment is similar to the micro light emitting diode structure **200***a* of FIG. **10**, and a difference there between is that in the embodiment, in cross-section, the outline of the bonding layer **250***c* is a flat cap shape, but the invention is not limited thereto. Furthermore, a ration of a maximum radius R**1** of the upper surface **252***c* to a maximum radius R**2** of a bottom surface **254***c* of the bonding layer **250***c* in contact with the barrier layer **250***c* is between 0.5 and 1. An angle A between the upper surface **252***c* and the bottom surface **254***c* of the bonding layer **250***c* is less than or equal to 60 degrees.

(44) FIG. **13** is a schematic top view of a micro light emitting diode structure according to another embodiment of the invention. Referring to FIG. **10** and FIG. **13** at the same time, a micro light emitting diode structure **200***d* of the embodiment is similar to the micro light emitting diode structure **200***a* of FIG. **10**, and a difference there between is that in the embodiment, the first electrode **222***d* of the electrode layer **220***d* is located in the first opening **242***d* of the insulating

layer **240***d*, and the first barrier portion **232***d* of the barrier layer **230***d* is disposed in the first opening **242***d*, covers the first electrode **222***d* and is partially extended on the insulating layer **240***d*. The second electrode **224***d* of the electrode layer **220***d* is located in the second opening **244***d* of the insulating layer **240***d*, and the second barrier portion **234***d* of the barrier layer **230***d* is disposed in the second opening **244***d*, covers the second electrode **224***d* and is partially extended on the insulating layer **240***d*. Here, the first electrode **222***d* and the second electrode **224***d* are only located in the first opening **242***d* and the second opening **244***d*, respectively. The bonding layer **250***d* is disposed on the barrier layer **230***d* and away from the electrode layer **220***d*.

(45) FIG. **14** is a schematic top view of a micro light emitting diode structure according to another embodiment of the invention. Referring to FIG. **13** and FIG. **14** at the same time, a micro light emitting diode structure **200***e* of the embodiment is similar to the micro light emitting diode structure **200***d* of FIG. **13**, and a difference there between is that in the embodiment, an orthographic projection of the bonding layer **250***e* on the epitaxial structure **210** is less than and covered by an orthographic projection of the barrier layer **230***d* on the epitaxial structure **210**, and a ration of the orthographic projection of the bonding layer **250***e* to the orthographic projection of the barrier layer **230***d* is greater than or equal to 0.5 and less than 1 to prevent overflow.

(46) In brief, in the embodiment, referring to FIG. **10** and FIG. **14** at the same time, an orthographic projection of the bonding layer **250***a*, **250***e* on the epitaxial structure **210** is less than or equal to and covers an orthographic projection of the barrier layer **230***a*, **230***d* on the epitaxial structure **210**, and a ration of the orthographic projection of the bonding layer **250***a*, **250***e* to the orthographic projection of the barrier layer **230***a*, **230***d* is greater than or equal to 0.5 and less than or equal to 1. If the above ratio is too large, the bonding **250***a*, **250***e* will overflow; if the above ration is too small, the bonding force of the bonding layer **250***a*, **250***e* will be insufficient.

(47) FIG. **15** is a schematic cross-sectional view of a micro light emitting diode display device according to another embodiment of the invention. Referring to FIG. **15**, in the embodiment, the micro light emitting diode display device **30** includes a display substrate **40** and, for example, a plurality of the micro light emitting diode structure **200***a* shown in FIG. **10**, where the micro light emitting diode structures **200***a* are disposed on the display substrate **40**. In detail, in the embodiment, the display substrate **40** includes a plurality of pads **42**, and the bonding layer **250***a* of each of micro light emitting diode structure **200***a* is bonded to the pads **22**, so that the micro light emitting diode structure **200***a* is disposed on the display substrate **40** and is electrically connected to the display substrate **40**. Here, the bonding layer **250***a* covers the barrier layer **230***a* and directly contacts the insulating layer **240***a*. The barrier layer **230***a* is disposed on the electrode layer **220***a*, and the bonding layer **250***a* is disposed on the barrier layer **230***a* and away from the electrode layer **220***a*, which may effectively avoid a problem that the bonding layer **250***a* overflows to form a eutectic with the first electrode **222***a* and the second electrode **224***a* of the electrode layer **220***a* due to a high temperature and high pressure process of eutectic bonding when the micro light emitting diode structure **200***a* is bonded to the display substrate **40**. In addition to preventing the eutectic problem between the electrode layer **220***a* and the bonding layer **250***a*, the barrier layer **230***a* can also enable the bonding layer **250***a* disposed thereon to have batter adhesion.

(48) It should be noted that in other embodiments that are not shown, the micro light emitting diode display device may include a plurality of any of the micro light emitting diode structures **200***a*, **200***b*, **200***c*, **200***d*, **200***e* according to actual requirements, which is not limited by the invention. Furthermore, in one embodiment, the bonding layer can also be provided on the pads to increase the bonding yield. In addition, the micro light emitting diode structure may be a red micro light emitting diode structure, a blue micro light emitting diode structure or a green micro light emitting diode structure. Moreover, the display substrate **40** of the embodiment may be, for example, a complementary metal-oxide-semiconductor (CMOS) substrate, a liquid crystal on silicon (LCOS) substrate, a thin film transistor (TFT) substrate or other substrates with working circuits, which are not limited by the invention.

(49) In summary, in the design of the micro light emitting diode structure of the invention, the barrier layer is disposed on the electrode layer, and the bonding layer is disposed on the barrier layer and away from the electrode layer. Namely, in the invention, the barrier layer is used to prevent the eutectic problem between the electrode layer and the bonding layer, and can make the bonding layer disposed on the barrier layer have better adhesion. Therefore, the micro light emitting diode structure of the invention may have better electrical properties and structural reliability, and the micro light emitting diode display device using the micro light emitting diode structures of the invention may have better display yield.
(50) It will be apparent to those skilled in the art that various modifications and variations can be made to the disclosed embodiments without departing from the scope or spirit of the invention. In view of the foregoing, it is intended that the invention covers modifications and variations provided they fall within the scope of the following claims and their equivalents.

Claims

1. A micro light emitting diode structure, comprising: an epitaxial structure, comprising a first-type semiconductor layer, a light emitting layer and a second-type semiconductor layer, and having a surface and a through hole, wherein the light emitting layer is located between the first-type semiconductor layer and the second-type semiconductor layer, the first-type semiconductor layer has the surface, the through hole penetrates through the first-type semiconductor layer, the light emitting layer and a part of the second-type semiconductor layer, and the first-type semiconductor layer, the light emitting layer and the second-type semiconductor layer surround the through hole; an electrode layer, disposed on the surface of the epitaxial structure and in the through hole of the epitaxial structure, wherein the electrode layer directly contacts the epitaxial structure, and the electrode layer has an upper surface relatively far away from the epitaxial structure and a peripheral surface connected to the upper surface; a barrier layer, directly disposed on the electrode layer, wherein the barrier layer completely covers the upper surface and the peripheral surface of the electrode layer; and a bonding layer, disposed on the barrier layer and away from the electrode layer.
2. The micro light emitting diode structure as claimed in claim 1, wherein the electrode layer comprises a first electrode and a second electrode, the barrier layer comprises a first barrier portion and a second barrier portion, the first barrier portion covers the first electrode, the second barrier portion covers the second electrode.
3. The micro light emitting diode structure as claimed in claim 2, further comprising: an insulating layer, disposed on the epitaxial structure and covering the surface and a peripheral surface of the epitaxial structure, wherein the insulating layer has a first opening exposing the first-type semiconductor layer and a second opening exposing the second-type semiconductor layer, the first electrode is disposed in the first opening and is electrically connected with the first-type semiconductor layer, and the second electrode is disposed in the second opening and is electrically connected with the second-type semiconductor layer.
4. The micro light emitting diode structure as claimed in claim 3, wherein the first barrier portion contacts the insulating layer and the first electrode, and the second barrier portion contacts the insulating layer and the second electrode, ratios of orthographic projections of the first electrode and the second electrode on the epitaxial structure to orthographic projections of the first barrier portion and the second barrier portion on the epitaxial structure are respectively greater than or equal to 0.25 and less than or equal to 0.95.
5. The micro light emitting diode structure as claimed in claim 3, wherein the first electrode is located in the first opening, the first barrier portion is disposed in the first opening, covers the first electrode and is partially extended on the insulating layer, the second electrode is located in the second opening, the second barrier portion is disposed in the second opening, covers the second

electrode and is partially extended on the insulating layer.

6. The micro light emitting diode structure as claimed in claim 1, wherein an orthographic projection of the bonding layer on the epitaxial structure is less than or equal to and covers an orthographic projection of the barrier layer on the epitaxial structure, and a ration of the orthographic projection of the bonding layer to the orthographic projection of the barrier layer is greater than or equal to 0.5 and less than or equal to 1.

7. The micro light emitting diode structure as claimed in claim 1, wherein the bonding layer has an upper surface, and the upper surface is a convex curved surface.

8. The micro light emitting diode structure as claimed in claim 7, wherein a curvature of the upper surface gradually decreases from a portion close to the barrier layer to another portion away from the barrier layer.

9. The micro light emitting diode structure as claimed in claim 7, wherein a ration of a maximum radius of the upper surface to a maximum radius of a bottom surface of the bonding layer in contact with the barrier layer is between 0.5 and 1.

10. The micro light emitting diode structure as claimed in claim 9, wherein an angle between the upper surface and the bottom surface of the bonding layer is less than or equal to 60 degrees.

11. The micro light emitting diode structure as claimed in claim 1, wherein the bonding layer has a maximum thickness at a center and an edge thickness at an edge, and the maximum thickness is greater than twice the edge thickness.

12. The micro light emitting diode structure as claimed in claim 1, wherein there is a distance between an edge of the bonding layer and an edge of the electrode layer, and a ration of an orthographic projection of the distance on the epitaxial structure to an orthographic projection of the electrode layer on the epitaxial structure is greater than or equal to 0.2.

13. A micro light emitting diode display device, comprising: a display substrate, comprising a plurality of pads; and a plurality of micro light emitting diode structures, disposed on the display substrate and respectively comprising: an epitaxial structure, comprising a first-type semiconductor layer, a light emitting layer and a second-type semiconductor layer, and having a surface and a through hole, wherein the light emitting layer is located between the first-type semiconductor layer and the second-type semiconductor layer, the first-type semiconductor layer has the surface, the through hole penetrates through the first-type semiconductor layer, the light emitting layer and a part of the second-type semiconductor layer, and the first-type semiconductor layer, the light emitting layer and the second-type semiconductor layer surround the through hole; an electrode layer, disposed on the surface of the epitaxial structure and in the through hole of the epitaxial structure, wherein the electrode layer directly contacts the epitaxial structure, and the electrode layer has an upper surface relatively far away from the epitaxial structure and a peripheral surface connected to the upper surface; a barrier layer, directly disposed on the electrode layer, wherein the barrier layer completely covers the upper surface and the peripheral surface of the electrode layer; and a bonding layer, disposed on the barrier layer, wherein the bonding layer is bonded to the corresponding pads, so that the micro light-emitting diode structures are electrically connected to the display substrate.

14. The micro light emitting diode display device as claimed in claim 13, wherein each of the plurality of micro light emitting diode structures further comprises: an insulating layer, disposed on the epitaxial structure and covering the surface and a peripheral surface of the epitaxial structure, wherein the insulating layer has a first opening exposing the first-type semiconductor layer and a second opening exposing the second-type semiconductor layer, the electrode layer comprises a first electrode and a second electrode, the first electrode is disposed in the first opening and is electrically connected with the first-type semiconductor layer, and the second electrode is disposed in the second opening and is electrically connected with the second-type semiconductor layer, and the bonding layer directly contacts the insulating layer.